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PART 1: PROFILE OF REPORTING BODY

1(a) Name of reporting body
Glasgow City Council

1(b) Type of body
Local Government

1(c) Highest number of full-time equivalent staff in the body during the report year
16612.3

1(d) Metrics used by the body			
Specify the metrics that the body uses to assess its performance in relation to climate change and sustainability.			
Metric	Unit	Value	Comments
Floor area	m2	1387719	Total Internal floor area
Population size served	population	606340	Total population in the Local Authority area

1(e) Overall budget of the body	
Specify approximate £/annum for the report year.	
Budget	Budget Comments
1344000000	

1(f) Report year	
Specify the report year.	
Report Year	Report Year Comments
Financial (April to March)	

1(g) Context
Provide a summary of the body's nature and functions that are relevant to climate change reporting.
Glasgow City Council is the unitary local government authority for the city and hosts the Sustainable Glasgow multi-sectoral partnership, which is leading on climate change policy. The Leader of the City Council also chairs Sustainable Glasgow. The Council has taken a key role in the regional adaptation initiative Climate Ready Clyde and it provided funding towards the early development of the partnership.
The Council's senior management and elected members backed a successful climate-change-themed bid for the Rockefeller Foundation's 100 Resilient Cities award. The Council consulted on its draft Resilience Strategy during the reporting period, with climate change adaptation being a key aspect of it. The strategy was formally launched in September 2016.
The Commonwealth Games Greener Legacy programme, including projects with benefits for mitigation and community-based adaptation, has been led and promoted by the Depute Leader of the council. In 2015, Glasgow celebrated its first ever Green Year with a series of thematic monthly events to communicate and engage with staff and communities on issues relevant to Climate Change Adaptation and Mitigation.

PART 2: GOVERNANCE, MANAGEMENT AND STRATEGY

2(a) How is climate change governed in the body?
Provide a summary of the roles performed by the body's governance bodies and members in relation to climate change. If any of the body's activities in relation to climate change sit outside its own governance arrangements (in relation to, for example, land use, adaptation, transport, business travel, waste, information and communication technology, procurement or behaviour change), identify these activities and the governance arrangements.
The diagram below outlines the governance of climate change mitigation and adaptation initiatives. These are fed into the / the Carbon Management Board and/or the Sustainable Glasgow Boardf. Sustainable Glasgow is chaired by the leader of the council.
Formal reporting is provided to the Environment and Sustainability Policy Development Committee and ultimately the Executive Committee for strategic decisions. Committees include elected members form all political groups who provide an active role in commenting on decisions.

2(b) How is climate change action managed and embedded by the body?
Provide a summary of how decision-making in relation to climate change action by the body is managed and how responsibility is allocated to the body's senior staff, departmental heads etc. If any such decision-making sits outside the body's own governance arrangements (in relation to, for example, land use, adaptation, transport, business travel, waste, information and communication technology, procurement or behaviour change), identify how this is managed and how responsibility is allocated outside the body (JPEG, PNG, PDF, DOC)
The attached diagram outlines the internal management structure for internal decision making. The Executive Director of Land & Environmental Services is the Green Champion. This work is driven by the Head of Sustainability and various workstreams including Sustainable Glasgow, Resilient Glasgow, City Energy and Carbon Management. Specifically the Climate Change officer sits in the Environment and Strategy team.
The Head of Sustainability (who is also the city's Chief Resilience Officer) chairs a number of internal meetings to discuss strategic decisions across the City Council in relation to climate change mitigation and/or adaptation. These include the Carbon Management Board, Resilient Glasgow steering group and the Senior Officer's group on Sustainability and the Environment (this includes representatives from all services and the wider Council family).
Glasgow City Council use the CCAT tool to facilitate interdepartmental engagement and assessment. The last workshop was run in June 2016 and an action plan is being prioritised for delivery during 2016/17 and review in 2017.

2(c) Does the body have specific climate change mitigation and adaptation objectives in its corporate plan or similar document?		
Provide a brief summary of objectives if they exist.		
Objective	Doc Name	Doc Link
'A Sustainable City' is one of five headline priority areas. Within this (p14-15), the four action themes include 'Reduced carbon footprint', 'Increased use of public and green transport', 'Connected transport infrastructure' and ' Accessible and affordable Transport'.	Glasgow City Council's Refreshed Strategic Plan for 2015-17	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=30229&p=0

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2(c) Does the body have specific climate change mitigation and adaptation objectives in its corporate plan or similar document?		
Provide a brief summary of objectives if they exist.		
Objective	Doc Name	Doc Link
Of the 15 Actions within the Air Quality Action Plan, some address transport aspects of adaptation. Many address mitigation explicitly or implicitly, notably: A Idling Vehicles, D Council Workplace Travel Planning, F Boiler Emissions (energy efficiency), H Leading by Example (Council vehicle fleet), M fire reduction, N Tree Planting.	Air Quality Action Plan 2009	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=32447&p=0
the 'Sustainable City' priority area includes a particular focus on reducing emissions and on adaptation policy and strategy. Section 2/Appendix 2 – Details commitments under “the sustainable city” theme for 2016/17. Appendix 4 – details other service priorities for 2016/17 some of which are areas such as Renewable Energy Projects and LED Lighting project. Appendix 5 - details 2015/16 performance against the priorities for that year under the sustainable city theme and our other operational priorities.	Service Plan - Land & Environmental Services	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=34173&p=0
In the proposed LDP, the City Development Plan: - flooding is named as a key challenge/influence on land-use planning (p14), and adaptation (mainly of the natural environment) features in the Green Place Strategic Outcome (p23) - in the two overarching policies, Placemaking Principle (CDP1) refers to mitigation and adaptation, and Sustainable Spatial Strategy (CDP2) refers to adaptation - mitigation is presented as the key context for the Resource Management policy (CDP5) - adaptation is one of many functions required by the Green Network policy (CDP6), and is presented as the key context for the Water Environment policy (CDP8) In the adopted City Plan 2: - - the Guiding Principles include sustainable development, with climate change (mitigation and adaptation) a key component. Adaptation is specifically mentioned in the policy on drainage & flooding (ENV4, ENV5) and infrastructure (e.g. River Clyde Flood Management Strategy Design Guide). Mitigation is specifically mentioned in several policies, principally sustainable design and energy (DES1, DES2, ENV15).	City Development Plan	http://connect.glasgow.gov.uk/CHttpHandler.ashx?id=28131&p=0 https://www.glasgow.gov.uk/index.aspx?articleid=16185
The key trends identified include adaptation (city greening) and mitigation (benefits of city density and ease of accessibility). Adaptation implicitly pervades the Districts Strategy and its placemaking focus on retrofitted 'Avenues', including active travel and integrated green infrastructure. Green infrastructure and reduced waste and emissions form Priority Actions under the respective themes of Infrastructure and Management.	City Centre Strategy 2014-19	https://www.glasgow.gov.uk/index.aspx?articleid=18277

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2(d) Does the body have a climate change plan or strategy?
If yes, provide the name of any such document and details of where a copy of the document may be obtained or accessed.
The Energy and Carbon Masterplan (ECMP) was published in 2014. The ECMP sets out a blueprint for how Glasgow can reduce its carbon dioxide emissions by 30% by 2020. The Energy and Carbon Masterplan presents over 30 actions that aim to meet the 30% reduction target. The main actions focus on reducing the total energy the city uses and ensuring that the city uses energy efficiently. Coupled with this, a series of actions aim to increase the proportion of energy consumed from low carbon and renewable sources. The expansion of wind energy, solar photovoltaic panels, geothermal energy and energy recovery from waste are all technologies that can be used in Glasgow - generating cheaper, clean energy and creating jobs.

2(e) Does the body have any plans or strategies covering the following areas that include climate change?				
Provide the name of any such document and the timeframe covered.				
Topic area	Name of document	Link	Time period covered	Comments
Adaptation				Glasgow City Council currently does not have a stand-alone Climate Adaptation Strategy. However, as a signatory to the EU Mayors Adapt, the Council is working towards embedding climate adaptation across its corporate policies. The Council is continuing its involvement in Climate Ready Clyde Partnership and pursuing the development of a Regional Climate Adaptation Strategy.
Business travel	Carbon Management Plan	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=28416&p=0	2013-2021	Climate Change is one of the main drivers identified in the Carbon Management Plan. The plan includes Business travel (pp 25), Transport is identified as being 'within scope'.The data for transport includes all council fleet vehicles, including cars, vans and refuse vehicles; and vehicle mileage used in the course of council business, including personal vehicles, hire cars and taxis.
Staff Travel	Staff Travel Plan		On-going	The Council are committed to Scotland's Climate Change Declaration, working to reduce CO2 emissions through the Carbon Management Plan. The Staff Travel Plan is a key way to achieve this and reduce carbon emissions.

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2(e) Does the body have any plans or strategies covering the following areas that include climate change?				
Provide the name of any such document and the timeframe covered.				
Topic area	Name of document	Link	Time period covered	Comments
Energy efficiency	Carbon Management Plan	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=28416&p=0	2013-2021	Climate Change is one of the main drivers identified in the Carbon Management Plan. The Carbon management Team leads the plan. The team is responsible for identifying, monitoring, reporting and supporting projects that reduce all emissions relating to our core services and those of our ALEOs.
Fleet transport	Carbon Management Plan	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=28416&p=0	2013-2021	The plan includes Fleet transport as well as business travel (pp 25), Transport is identified as being 'within scope.'The data for transport includes all council fleet vehicles, including cars, vans and refuse vehicles; and vehicle mileage used in the course of council business, including personal vehicles, hire cars and taxis.
Information and communication technology	Carbon Management Plan	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=28416&p=0	2013-2021	The Carbon Management Plan includes implementation of intelligent software programme which is not specifically targeted at ICT, but will have a positive impact in managing ICT devices (power off etc).
Renewable energy	Energy and Carbon Masterplan	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=32441&p=0	2014-2020	'The Energy and Carbon Masterplan presents over 30 actions that aim to meet the 30% reduction target. The main actions focus on reducing the total energy the city uses and ensuring that the city uses energy efficiently. Coupled with this, actions aim to increase the proportion of energy consumed from low carbon and renewable sources.'
Sustainable/renewable heat	Energy and Carbon Masterplan	https://www.glasgow.gov.uk/CHttpHandler.ashx?id=32441&p=0	2014 - 2020	'The expansion of wind energy, solar photovoltaic panels, geothermal energy and energy recovery from waste are all technologies that can be used in Glasgow - generating cheaper, clean energy and creating jobs.'
Waste management	Tackling Glasgow's Waste, Cleansing Waste Strategy & Action Plan	https://www.glasgow.gov.uk/index.aspx?articleid=16572	2015-2020	Does not specifically mention climate change, but will contribute in a positive way
Water and sewerage	Metropolitan Glasgow Strategic Drainage Partnership (MGSDP)	https://www.mgsdp.org/index.aspx?articleid=2009	2060	Adaptation pervades the Partnership's Guiding Principles, which include 'Climate change ready'. However, this work has not yet been subjected to formal adaptation planning. The partnership published their 60 year vision, it states: 'The MGSDP 2060 Vision is to transform how the city region thinks about and manages rainfall to end uncontrolled flooding and improve water quality.'

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2(e) Does the body have any plans or strategies covering the following areas that include climate change?				
Provide the name of any such document and the timeframe covered.				
Topic area	Name of document	Link	Time period covered	Comments
Land Use	Local Development Plan (proposed)	https://www.glasgow.gov.uk/index.aspx?articleid=16186	to be adopted in late 2016	Flooding is named as a key challenge/influence on land-use planning (p14), and adaptation (mainly of the natural environment) features in the Green Place Strategic Outcome (p23) In the two overarching policies, Placemaking Principle (CDP1) refers to mitigation and adaptation, and Sustainable Spatial Strategy (CDP2) refers to adaptation Mitigation is presented as the key context for the Resource Management policy (CDP5) Adaptation is one of many functions required by the Green Network policy (CDP6), and is presented as the key context for the Water Environment policy (CDP8)
Other (state topic area covered in comments)	Local Development Plan	https://www.glasgow.gov.uk/index.aspx?articleid=16186	to be adopted in late 2016	The Sustainable Transport policy (CDP11) states that "The Council has a duty to act in a way considered to be most sustainable and best calculated to deliver reductions in greenhouse gas emissions. Reducing the need to travel, particularly by car, is one of the key ways in which the City can help reduce emissions of greenhouse gases."
Land Use	Local Development Plan	https://www.glasgow.gov.uk/index.aspx?articleid=16186	to be adopted in late 2016	Supplementary Guidance on Water Environment (paragraph 4, 2 and 14.2) mention climate adaptation, placemaking and green infrastructure specifically The Guiding Principles include sustainable development, with climate change (mitigation and adaptation) a key component. Mitigation is specifically mentioned in several policies, principally sustainable design and energy (DES1, DES2, ENV15). Adaptation is specifically mentioned in the policy on drainage & flooding (ENV4, ENV5) and infrastructure (e.g. River Clyde Flood Management Strategy Design Guide). Climate Adaptation & Integrated Green Infrastructure is one of the 5 key themes within the forthcoming Open Space Strategy.Delivery of multi-functional open space is fundamental to the strategy, with direct links to placemaking, health and wellbeing, climate adaptation and mitigation.

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2(f) What are the body’s top 5 priorities for climate change governance, management and strategy for the year ahead?
Provide a brief summary of the body's areas and activities of focus for the year ahead.
Sustainable Glasgow is the lead body for driving the city’s adaptation agenda forward. Its workplan schedules presentations and reports to the Board on adaptation, linking closely to the city’s engagement in the Climate Ready Clyde regional partnership. The chair of Sustainable Glasgow will announce adaptation progress to the 2017 ECCA conference Publish a city wide Resilience Strategy - This strategy is the key initial outcome of Glasgow's selection in 2013 as one the Rockefeller Foundation's 100 Resilient Cities. Develop set of indicators to monitor progress of the Resilience Strategy. Establish internal and external working/governance for preparing a city-wide Climate Adaptation Framework, building on the recommendations from the Resilience Strategy and in alignment with the climate Ready Clyde Partnership's vision and objectives. Continue to take active part in the Climate Ready Clyde Partnership.

2(g) Has the body used the Climate Change Assessment Tool(a) or equivalent tool to self-assess its capability / performance?
If yes, please provide details of the key findings and resultant action taken.
Yes, the council has used the CCAT tool recently and plans to have a CACT workshop annually to assess progress against the proposed action plan. The last workshop was well attended and included representatives from across the different sections or Climate Action. It raised interesting discussions around the council's strengths and weaknesses in the context of climate change action. It also highlighted the need to continue to raise awareness of Climate Action and public bodies' duties as well as clarifying the role that each service area plays in addressing climate change. The results show that the council scored highly for Emissions which reflects the commitment of the Corporate Management Team. The council has put carbon management (Climate Change Mitigation) at the core of the key objectives and senior management are accountable for achieving targets and implementing plans. Data collection systems are effective, allowing good tracking of the carbon footprint and the project list is well documented, although it required widen its scope to include areas such as waste to enable to carbon reduction target to be met. Carbon reduction targets and associated projects are reviewed regularly and it is proposed that an annual CACT review is used to widen the scope of the performance improvement process. The exercise indicated that there is room for improvement in the Council’s approach to adaptation and incorporating it in its broader risk assessment framework. These issues are being addressed within the Council and as part of its membership of the Climate Ready Clyde Partnership. Internal communication and staff engagement are also areas highlighted as requiring improvement. The successes of the Carbon Management Team are communicated frequently and effectively, this should be widened to include other areas of climate action. It should pave the way for better staff engagement and bottom up activity to identify and implement climate change adaptation and mitigation actions across the organisation. Procurement is an area where some progress has been made towards sustainability considerations being embedded across the procurement process and policy, especially around carbon reduction and energy efficiency. There is still a way to go to fully incorporate climate change adaptation considerations in procurement. The resulting action plan is being circulated to all attendees and the group are still to agree on the 10 top priority actions for the year ahead.

2(h) Supporting information and best practice
Provide any other relevant supporting information and any examples of best practice by the body in relation to governance, management and strategy.
Sustainable Glasgow is the key partnership for delivering on city-wide sustainability aims, bringing together public bodies, the private sector and academia. It is chaired by the Leader of the Council, thereby ensuring the Council’s strong leadership role in progressing the mitigation and adaptation agendas. The partnership has driven a number of green energy initiatives with a focus on affordable warmth and emissions reduction targets. This work has aimed to deliver major investment, help regenerate communities and tackle fuel poverty, and improve environmental quality. The Sustainable Glasgow Board continues to be committed to addressing climate change adaptation and agreed to establish a workstream to progress it within the Council and across partners. The Board receives regular update reports on adaptation, alongside those presented to Council committees.

PART 3: EMISSIONS, TARGETS AND PROJECTS

3a Emissions from start of the year which the body uses as a baseline (for its carbon footprint) to the end of the report year							
Complete the following table using the greenhouse gas emissions total for the body calculated on the same basis as for its annual carbon footprint /management reporting or, where applicable, its sustainability reporting. Include greenhouse gas emissions from the body's estate and operations (a) (measured and reported in accordance with Scopes 1 & 2 and, to the extent applicable, selected Scope 3 of the Greenhouse Gas Protocol (b)). If data is not available for any year from the start of the year which is used as a baseline to the end of the report year, provide an explanation in the comments column. (a) No information is required on the effect of the body on emissions which are not from its estate and operations.							
Reference Year	Year	Scope1	Scope2	Scope3	Total	Units	Comments
Baseline carbon footprint	2005/06				199919	tCO2e	
Year 1 carbon footprint	2006/07				199919	tCO2e	Emissions were not recorded in 2006/27 hence the same value as for 2005/06
Year 2 carbon footprint	2007/08				179355	tCO2e	
Year 3 carbon footprint	2008/09				182786	tCO2e	
Year 4 carbon footprint	2009/10				186359	tCO2e	
Year 5 carbon footprint	2010/11				186878	tCO2e	
Year 6 carbon footprint	2011/12				171225	tCO2e	
Year 7 carbon footprint	2012/13				189901	tCO2e	
Year 8 carbon footprint	2013/14				179397	tCO2e	
Year 9 carbon footprint	2014/15				175968	tCO2e	
Year 10 carbon footprint	2015/16				171220	tCO2e	

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3b Breakdown of emission sources									
Complete the following table with the breakdown of emission sources from the body's most recent carbon footprint (greenhouse gas inventory); this should correspond to the last entry in the table in 3 (a) above. Use the 'Comments' column to explain what is included within each category of emission source entered in the first column. If, for any such category of emission source, it is not possible to provide a simple emission factor(a) leave the field for the emission factor blank and provide the total emissions for that category of emission source in the 'Emissions' column.									
Total	Comments – reason for difference between Q3a & 3b.	Emission source	Scope	Consumption data	Units	Emission factor	Units	Emissions (tCO2e)	Comments
135670.8	2014/15 factors have been employed in 2015/16 to allow for more accurate comparison of performance.	Grid Electricity (generation)	Scope 2	123134402	kWh	0.46219	kg CO2e/kWh	56911.5	
		Grid Electricity (transmission & distribution losses)	Scope 2	123134402	kWh	0.03816	kg CO2e/kWh	4698.8	
		Natural Gas	Scope 1	274731280	kWh	0.18445	kg CO2e/kWh	50674.2	
		Diesel (average biofuel blend)	Scope 3	5655400	litres	2.58390	kg CO2e/litre	14613.0	
		Gas Oil	Scope 1	2362797	litres	2.90884	kg CO2e/litre	6873.0	
		Coal (industrial)	Scope 1	95	tonnes	2466.27400	kg CO2e/tonne	234.3	
		Petrol (average biofuel blend)	Scope 3	132181	litres	2.19440	kg CO2e/litre	290.1	
		Domestic flight (average passenger)	Scope 3	1326130	passenger km	0.29795	kg CO2e/passenger km	395.1	
		Rail (National rail)	Scope 3	602174	passenger km	0.04506	kg CO2e/passenger km	27.1	
		Average Car - Unknown Fuel	Scope 3	3180137	miles	0.299901254	kg CO2e/mile	953.7	

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3c Generation, consumption and export of renewable energy					
Provide a summary of the body's annual renewable generation (if any), and whether it is used or exported by the body.					
	Renewable Electricity		Renewable Heat		
Technology	Total consumed by the organisation (kWh)	Total exported (kWh)	Total consumed by the organisation (kWh)	Total exported (kWh)	Comments
Solar PV	9459	9459			Deemed supply so installations assumed to export 50% of total generated electricity
Biomass			3004390	0	

3d Targets										
List all of the body's targets of relevance to its climate change duties. Where applicable, overall carbon targets and any separate land use, energy efficiency, waste, water, information and communication technology, transport, travel and heat targets should be included.										
Name of Target	Type of Target	Target	Units	Boundary/scope of Target	Progress against target	Year used as baseline	Baseline figure	Units of baseline	Target completion year	Comments
2020 Carbon Reduction Target	absolute	30	total % reduction	Other (please specify in comments)	14.4	2005/06	199919	tCO2e	2020	All measured emissions including buildings, streetlights, transport, staff travel

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3e Estimated total annual carbon savings from all projects implemented by the body in the report year			
Total	Emissions Source	Total estimated annual carbon savings (tCO2e)	Comments
2689	Electricity	1400	Combination of LED lighting, VSDs and Voltage Optimisation
	Natural gas	1289	Predominantly heating control in the form of BMS installs/retrofit
	Other heating fuels		
	Waste		
	Water and sewerage		
	Business Travel		
	Fleet transport		
	Other (specify in comments)		

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3f Detail the top 10 carbon reduction projects to be carried out by the body in the report year											
Provide details of the 10 projects which are estimated to achieve the highest carbon savings during report year.											
Project name	Funding source	First full year of CO2e savings	Are these savings figures estimated or actual?	Capital cost (£)	Operational cost (£/annum)	Project lifetime (years)	Primary fuel/emission source saved	Estimated carbon savings per year (tCO2e/annum)	Estimated costs savings (£/annum)	Behaviour Change	Comments
RSBi Heating Upgrade	Capital and CEEF	2015	Estimated	300000		20	Natural Gas	203	34500		Replacement of old boilers with high efficiency CHP and controls.
Shawlands High School LED Retrofit	CEEF	2016	Estimated	224519	0	10	Grid Electricity	192	42538		Completed over two phases. Phase 1 savings verified, phase 2 savings verification pending.
Duke St Car Park lighting control	CEEF	2016	Actual	34828	0	10	Grid Electricity	75	17660		
Blochairn Fruitmarket LED Replacement	CEEF	2016	Actual	59706	0	10	Grid Electricity	64	14776		
Queenspark Depot LED retrofit	CEEF	2015	Estimated	6250	0	10	Grid Electricity	43	9683		
The Lighthouse installation of VSDs on HVAC plant	CEEF	2015	Actual	7900		10	Grid Electricity	41	9163		
Upgrade of heating control in 5 parks depots	CEEF	2016	Estimated	21120		10	Grid Electricity	128	29009		Locked timers on electric radiators
The Lighthouse Upgrade of BMS	CEEF	2016	Actual	53353		10	Natural Gas	35	5659		
The Lighthouse VSD and controls on escalators	CEEF	2016	Actual	30475		10	Grid Electricity	31	6463		
Installation of pool covers at 8 Secondary Schools	Capital	2016	Estimated	126071		5	Other	202	34737		Combined savings relating to both gas and electricity consumption reduction

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3g Estimated decrease or increase in the body's emissions attributed to factors (not reported elsewhere in this form) in the report year				
If the emissions increased or decreased due to any such factor in the report year, provide an estimate of the amount and direction.				
Total	Emissions source	Total estimated annual emissions (tCO2e)	Increase or decrease in emissions	Comments
	0 Estate changes			This is not captured in any meaningful way in relation to the carbon management programme.
	Service provision			This is not captured in any meaningful way in relation to the carbon management programme.
	Staff numbers			This is not captured in any meaningful way in relation to the carbon management programme.
	Other (specify in comments)			

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3h Anticipated annual carbon savings from all projects implemented by the body in the year ahead			
Total	Source	Saving	Comments
1201	Electricity	733	As of August 2016
	Natural gas	468	As of August 2016
	Other heating fuels		
	Waste		
	Water and sewerage		
	Business Travel		
	Fleet transport		
	Other (specify in comments)		

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3i Estimated decrease or increase in the body's emissions attributed to factors (not reported elsewhere in this form) in the year ahead				
If the emissions are likely to increase or decrease due to any such factor in the year ahead, provide an estimate of the amount and direction.				
Total	Emissions source	Total estimated annual emissions (tCO2e)	Increase or decrease in emissions	Comments
0	Estate changes		Decrease	This is not captured in any meaningful way in relation to the carbon management programme, however the implementation of Transformation Programmes allows a general assumption of reduction in the number of buildings within the estate.
	Service provision			This is not captured in any meaningful way in relation to the carbon management programme.
	Staff numbers		Decrease	This is not captured in any meaningful way in relation to the carbon management programme, however the implementation of Transformation Programmes allows a general assumption of reduction in the number of staff.
	Other (specify in comments)			

3j Total carbon reduction project savings since the start of the year which the body uses as a baseline for its carbon footprint	
If the body has data available, estimate the total emissions savings made from projects since the start of that year ("the baseline year").	
Total	Comments
176452	This represents the total cumulative avoided emissions since baseyear as a result of Council activity, be it projects, building rationalisation or general efficiencies.

3k Supporting information and best practice
Provide any other relevant supporting information and any examples of best practice by the body in relation to its emissions, targets and projects.
Consideration is currently being given to methods of assessing the lifecycle costs of buildings in terms of their carbon costs and construction value.

PART 4: ADAPTATION

4(a) Has the body assessed current and future climate-related risks?
If yes, provide a reference or link to any such risk assessment(s). a)Current Risks A semi-formal assessment of current climate threats and opportunities has been undertaken with the Council's Parks & Open Spaces team. With regard to the natural environment, this work is considering how to make allowance for climate-related habitat change in various types of open space. Damage to buildings and infrastructure was identified as the dominant recent weather impact in the Council's Local Climate Impacts Profile (Dec 2011). Significant societal impacts were also identified. The work of the Council's Resilience team includes planning for weather-related incidents. It has not yet taken account of climate change predictions. Work to map current climate risk against community vulnerability has been undertaken by Glasgow & Clyde Valley Green Network Partnership on behalf of the Council, to inform the Glasgow Open Space Strategy. b) Future Risks A semi-formal assessment of future climate threats and opportunities has been undertaken with the Council's Parks & Open Spaces team .This work is identifying threats and opportunities for the natural environment, buildings / infrastructure and people within various types of open space. It has not yet directly looked at how greenspace contributes to addressing wider threats and opportunities in the city. Similar exercises were undertaken with roads and transport planning, city energy, public health and waste management, and the development plan team. These are still to be formalised and scoring to be agreed on. The Council has also begun the process of identifying high-level risks of future climate change for the organisation and is looking at engaging city wide stakeholders in assessing the main high-level climate risks for Glasgow, in line with UKCCRA risks identified for Scotland. This will be aligned with the Climate Ready Clyde initiative and its objectives.
4(b) What arrangements does the body have in place to manage climate-related risks?
Provide details of any climate change adaptation strategies, action plans and risk management procedures, and any climate change adaptation policies which apply across the body. Strategic responsibility for adaptation planning within the Council lies with the Environment & Strategy team in Land & Environmental Services, and specifically the Climate Change Officer. The approach being developed uses risk assessment methodology with individual service areas, building their capacity to devise adaptation measures for incorporation into their ongoing work-planning. Workshops have been held with the Parks & Open Spaces team and also with Environmental Health & Waste Management, City Energy, Development planning and Roads & Transport planning. High-level regional adaptation planning is undertaken within the Climate Ready Clyde initiative. Climate Ready Clyde is a Glasgow and Clyde valley wide regional partnership initiative which aims to produce a region wide adaptation strategy and action plan. The work of the Council's Resilience team includes planning for weather-related incidents. For flooding, this work incorporates SEPA's modelling and its allowance for climate change. Otherwise it is founded in annual Met Office projections, and has not yet taken account of climate change predictions. Within Development & Regeneration Services, the work of the Flood Risk Management Group incorporates hydrological climate change predictions. It is mostly focused on integrated green network solutions, under the Metropolitan Glasgow Strategic Drainage Partnership. Adaptation is the base of the Partnership's Guiding Principles, which include 'Climate change ready'. However, this work has not yet been subjected to formal adaptation planning. The Council's proposed Local Development Plan takes account of adaptation requirements at the strategic and policy level, based on national policy and guidance. The City Centre Strategy & Action Plan 2014-19 also has a key role in framing adaptation, given the high profile of the city centre. Formal adaptation planning was not involved in preparing these strategies, but there is room for it to inform various forthcoming pieces of LDP Supplementary Guidance.

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4(c) What action has the body taken to adapt to climate change?
Include details of work to increase awareness of the need to adapt to climate change and build the capacity of staff and stakeholders to assess risk and implement action.
Development plan policies (existing and in the proposed LDP) require new developments to: - Contribute to delivery/enhancement of multifunctional Green Network, as appropriate to the scale of development and the nature of existing open space. - Contribute to reducing flood risk, including avoiding any increase in surface-water run-off. The LDP is also a key element of the delivery of the city's new resilience strategy. The Sustainable Glasgow team has engaged with colleagues in Land and Environmental Services to raise awareness of climate change impacts on service delivery and business continuity. The Sustainable Glasgow team ran a CCAT Tool workshop with representatives from Development and Regeneration Services, Land and Environmental Services and Corporate Services (please see below). This was a useful session in raising awareness of climate change action and clarifying links to other policy areas within the council. The Council participated in the Adaptation Learning Exchange (ALE) and continue to take part in this forum to exchange experiences and build capacity to effectively address climate change risks and increase climate resilience in the city and council's services. The Council participated in the Mayors Adapt Twinning initiative along with Edinburgh City Council and the City of Stockholm. It was a good opportunity to share experiences of climate adaptation, including risk assessment, policy development, governance arrangements and the importance of partnership working. Glasgow City is participating with UN Habitat in a pilot project to test the Climate Action Guiding Principles launched by UN-Habitat at the COP 21. The Council is playing an active role in two major EU programmes on resilience planning, both of which address climate change issues. One is called SMR and is a Horizon-funded project and the other is an URBACT project which is being led by the City of Rotterdam. Together with its 100RC membership, these allow Glasgow to share its experience and to learn from other cities

4(d) Where applicable, what progress has the body made in delivering the policies and proposals referenced N1, N2, N3, B1, B2, B3, S1, S2 and S3 in the Scottish Climate Change Adaptation Programme(a) ("the Programme")?					
If the body is listed in the Programme as a body responsible for the delivery of one or more policies and proposals under the objectives N1, N2, N3, B1,B2, B3, S1, S2 and S3, provide details of the progress made by the body in delivering each policy or proposal in the report year. If it is not responsible for delivering any policy or proposal under a particular objective enter "N/A" in the 'Delivery progress made' column for that objective. (a) This refers to the programme for adaptation to climate change laid before the Scottish Parliament under section 53(2) of the Climate Change (Scotland) Act 2009 (asp 12) which currently has effect. The most recent one is entitled "Climate Ready Scotland: Scottish Climate Change Adaptation Programme" dated May 2014.					
Objective	Objective reference	Theme	Policy / Proposal reference	Delivery progress made	Comments
Understand the effects of climate change and their impacts on the natural environment.	N1	Natural Environment	N1-5	A baseline study of future sea-level rise on the Clyde is being supported by the Council through its participation in the Glasgow & Clyde Valley Strategic Development Plan partnership. A semi formal risk assessment for parks and open spaces was undertaken and identified risks and opportunities resulting from a changing climate to natural environment. This is now being used to inform the development of the open space strategy for Glasgow City.	

4(d) Where applicable, what progress has the body made in delivering the policies and proposals referenced N1, N2, N3, B1, B2, B3, S1, S2 and S3 in the Scottish Climate Change Adaptation Programme(a) ("the Programme")?					
If the body is listed in the Programme as a body responsible for the delivery of one or more policies and proposals under the objectives N1, N2, N3, B1,B2, B3, S1, S2 and S3, provide details of the progress made by the body in delivering each policy or proposal in the report year. If it is not responsible for delivering any policy or proposal under a particular objective enter "N/A" in the 'Delivery progress made' column for that objective. (a) This refers to the programme for adaptation to climate change laid before the Scottish Parliament under section 53(2) of the Climate Change (Scotland) Act 2009 (asp 12) which currently has effect. The most recent one is entitled "Climate Ready Scotland: Scottish Climate Change Adaptation Programme" dated May 2014.					
Objective	Objective reference	Theme	Policy / Proposal reference	Delivery progress made	Comments
Support a healthy and diverse natural environment with capacity to adapt.	N2	Natural Environment	N2-2	Local enhancements to individual greenspaces including the Multifunctional Greenspace Project (part of the city's Commonwealth Games Legacy) and the award-winning Stalled Spaces project.	
Sustain and enhance the benefits, goods and services that the natural environment provides.	N3	Natural Environment	N3-3	A Strategic Tree Planting Programme aims to mitigate against risk of net loss of trees / woodland, on a basis of ecosystem services rather than simply tree numbers.	
Understand the effects of climate change and their impacts on buildings and infrastructure networks.	B1	Buildings and infrastructure networks	B1-1	Awareness raising workshops and a semi formal risk assessment were undertaken with Roads and Transport planning and the Development Plan team. The LCLIP exercise also highlighted damage to infrastructure as one of the main impacts of current weather patterns. During dissemination of the LLCIP findings, the likelihood of this impact being exacerbated by future climate change was discussed and consideration of this was included in the risk assessment exercises.	
Provide the knowledge, skills and tools to manage climate change impacts on buildings and infrastructure.	B2	Buildings and infrastructure networks	B2-1	Awareness raising workshops were undertaken with Roads and Transport planning and the development Plan team. On-going discussions with representatives from Transport and Infrastructure are focussing on impacts of a changing climate to this sector and how the council can best prepare and adapt.	

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4(d) Where applicable, what progress has the body made in delivering the policies and proposals referenced N1, N2, N3, B1, B2, B3, S1, S2 and S3 in the Scottish Climate Change Adaptation Programme(a) ("the Programme")?					
If the body is listed in the Programme as a body responsible for the delivery of one or more policies and proposals under the objectives N1, N2, N3, B1,B2, B3, S1, S2 and S3, provide details of the progress made by the body in delivering each policy or proposal in the report year. If it is not responsible for delivering any policy or proposal under a particular objective enter "N/A" in the 'Delivery progress made' column for that objective. (a) This refers to the programme for adaptation to climate change laid before the Scottish Parliament under section 53(2) of the Climate Change (Scotland) Act 2009 (asp 12) which currently has effect. The most recent one is entitled "Climate Ready Scotland: Scottish Climate Change Adaptation Programme" dated May 2014.					
Objective	Objective reference	Theme	Policy / Proposal reference	Delivery progress made	Comments
Increase the resilience of buildings and infrastructure networks to sustain and enhance the benefits and services provided.	B3	Buildings and infrastructure networks	B3-3	Surface Water Management Planning is established for 17 areas across Glasgow City. This work is fully integrated with joint SEPA/Council work on city-wide flood risk assessment and mapping. Nearly half the SWMPs are ready to progress to implementation subject to funding / procurement. The Council's Resilience unit uses contingency planning and a warning system to allow city infrastructure to be prepared for severe weather events.	
Understand the effects of climate change and their impacts on people, homes and communities.	S1	Society	S1-1	As part of the council's membership of the Rockefeller Foundation, a Resilience Strategy was produced and will be launched in September 2016. The Strategy is centred around the theme of People Make Glasgow Resilient. The strategy is supported by four pillars , one of them being about empowering Glaswegians. In parallel to this and linked to the Climate Ready Clyde vision, a project to understand climate change and their impacts on people and communities was undertaken in the north of the city - Climate ready Lambhill.	
Increase the awareness of the impacts of climate change to enable people to adapt to future extreme weather events.	S2	Society	S2-2	Climate ready Lambhill built on the strong community resilience and civic participation already present in the area to look at impacts of a changing climate on the community and on existing challenges. The project has received national attention and is being used to communicate and raise awareness of climate change impacts, flooding being the main challenge for this area of the city. There is currently another climate resilience project being carried out at community level, in Sighthill - north of the city – called Weathering Change. This aims to look at climate change impacts and raise awareness in the community about them, whilst building community resilience to them.	

4(d) Where applicable, what progress has the body made in delivering the policies and proposals referenced N1, N2, N3, B1, B2, B3, S1, S2 and S3 in the Scottish Climate Change Adaptation Programme(a) ("the Programme")?					
If the body is listed in the Programme as a body responsible for the delivery of one or more policies and proposals under the objectives N1, N2, N3, B1,B2, B3, S1, S2 and S3, provide details of the progress made by the body in delivering each policy or proposal in the report year. If it is not responsible for delivering any policy or proposal under a particular objective enter "N/A" in the 'Delivery progress made' column for that objective. (a) This refers to the programme for adaptation to climate change laid before the Scottish Parliament under section 53(2) of the Climate Change (Scotland) Act 2009 (asp 12) which currently has effect. The most recent one is entitled "Climate Ready Scotland: Scottish Climate Change Adaptation Programme" dated May 2014.					
Objective	Objective reference	Theme	Policy / Proposal reference	Delivery progress made	Comments
Support our health services and emergency responders to enable them to respond effectively to the increased pressures associated with a changing climate.	S3	Society	S3-1	The Resilience unit's warning system allows information on forecast severe weather events to be targeted at communities. Ongoing and proposed development of local energy solutions (such as district heating) is increasing community resilience to weather-related energy supply problems.	

4(e) What arrangements does the body have in place to review current and future climate risks?	
Provide details of arrangements to review current and future climate risks, for example, what timescales are in place to review the climate change risk assessments referred to in Question 4(a) and adaptation strategies, action plans, procedures and policies in Question 4(b).	
a)Current Risks The semi formal risk assessments undertaken internally are to be reviewed as part of the city wide climate adaptation framework engagement and consultation. b) Future Risks A semi-formal assessment of future climate threats and opportunities has been undertaken with the Council's Parks & Open Spaces team .This work is identifying threats and opportunities for the natural environment, buildings / infrastructure and people within various types of open space. It has not yet directly looked at how greenspace contributes to addressing wider threats and opportunities in the city. Similar exercises were undertaken with roads and transport planning, city energy, public health and Waste management, development plan tea these are still to be formalised and scoring to be agreed on. This will again be reviewed as part of the city wide adaptation framework consultation process. it will take into account the most recent UKCCRA risks and more specifically those identified for Scotland.	

4(f) What arrangements does the body have in place to monitor and evaluate the impact of the adaptation actions?

Please provide details of monitoring and evaluation criteria and adaptation indicators used to assess the effectiveness of actions detailed under Question 4(c) and Question 4(d).

The Council is committed to running a CCAT Tool workshop with wide representation from Council services and ALEOs. This will enable progress and performance to be monitored against annual climate action plans proposed and targets to become climate ready until 2020.

Glasgow is participating with UN Habitat in a pilot project to test the Climate Action Guiding Principles launched by UN-Habitat at the COP 21. This will produce a report with feedback on progress and performance of the organisation in relation to its Climate Action, this will be used to monitor and improve climate action performance.

The Resilience Strategy contains specific actions on climate change adaptation planning which will form part of a forthcoming monitoring and evaluation framework for city resilience.

4(g) What are the body's top 5 priorities for the year ahead in relation to climate change adaptation?

Provide a summary of the areas and activities of focus for the year ahead.

To continue to progress Weathering Change and improve climate resilience in the city.

Embed climate risks in the Council's risk register.

To implement the resilience strategy's two year action plan.

To set up a task group in collaboration with local higher education institutions in the city and other key partners to address climate adaptation at a city scale, developing a city wide climate adaptation framework for Glasgow, ensuring that this is aligned with the Climate Ready Clyde Partnership.

To continue to play an active role in helping to progress Climate Ready Clyde initiative and its Regional Climate Adaptation strategy.

4(h) Supporting information and best practice
Provide any other relevant supporting information and any examples of best practice by the body in relation to adaptation.
Glasgow City Council are an active member of Steering group for the Climate Ready Clyde Partnership and continue to lead the partnership forward, along with other key partners in the Region. The Council are also in the process of setting up a task group to drive forward strategic adaptation work within the city and this is aligned with regional priorities and resilience strategy's actions and recommendations.

PART 5: PROCUREMENT

5(a) How have procurement policies contributed to compliance with climate change duties?
Provide information relating to how the procurement policies of the body have contributed to its compliance with climate changes duties. In the council's Sustainable Procurement Policy and action plan, October 2010, the council sets out the way procurement supports the objectives of economic growth and sustainability. It drives the three sustainable principles which are social, economic and environmental into all regulated procurement activities in the council. The delivery of the council's Corporate Procurement Strategy 2013 – 2015 (Strategy), which set out the procurement aims, objectives and key priorities for the two year period it covered, formed a significant element of last year's climate change submission and again influences and supports the basis of the procurement submission for FY15.16. The Strategy closely aligned to the council's strategic Plan 2012-2017 and reflected both national and local policies and priorities including our approach to fair trade, sustainability, corporate social responsibility and Co-operative Glasgow. It incorporated a comprehensive Strategy Performance Action Plan (action plan) that comprised of 68 actions, of which, 30% directly supported our continued drive to embed sustainability within the Corporate Procurement Unit's procurement process and procedures whilst also promoting the benefits of embedding sustainable procurement to the key stakeholders within the council. The progress of the Strategy was regularly reviewed via the council's Procurement Governance Structure, which includes the Strategic Asset Management Programme (SAMP), whose members are responsible for delivering strategic guidance on how procurement is delivered across the council and ensuring delivery of the actions outlined in the strategy. The undernoted highlights the key procurement sustainable actions and outcomes, achieved via the successful delivery of the strategy, which directly contribute to the climate change duties: In adherence with council policies and also in light of the Procurement Reform (Scotland) Act 2014 drive to embed sustainability across Scottish public sector procurement activity, the council's Corporate Procurement Unit participated in and directly influenced a number of national pilots, run by Scottish Government consultants, relating to the development of Sustainable Procurement Tools (Tools) i.e. Sustainable Test, Prioritisation Tool and the Flexible Framework Self Assessment Tool (FFSAT). The council has since embedded the Tools within the strategic procurement process to assist and ensure best value. Utilisation of the tools also supports compliance with the Sustainable Procurement Duty. The key tools utilised by procurement are as follows: The prioritisation tool was utilised to identify the sustainable opportunities and priority areas of focus for the Corporate Procurement Unit (CPU); based on their 18 month work plan. The output is incorporated , in the form of a graph, within the Corporate Procurement Strategy 2016 – 2018 and will continue to be reviewed / updated to inform where procurement apply our resources and effort over this period to assist in embedding relevant and proportionate sustainable requirements into contracts and encourage good practice. The Sustainable Test is fully embedded within our six stage strategic sourcing process to help embed relevant and proportionate sustainable requirements into contracts and encourage good practice and compliance with the Act. We have also adapted our procurement templates, including our procurement strategy template to not only include the output of the Sustainable Test, but to also capture how the strategy will be developed to influence any identified sustainable opportunities and risks. The Procurement strategy requires client (project Lead) / Service Department approval prior to progressing with the tender, which assists in ensuring wider ownership and support of sustainable procurement across the council. The council acknowledges the Flexible Framework Self- Assessment Tool (FFSAT) as one of the foundations in delivering the sustainable procurement duty under the Procurement Reform Act. In light of the revised FFSAT, the council refrained from any further activity on the existing FF action plan - concluded based on performance level3, with a view of progressing any further sustainable performance improvements via the completion of the revised FF and implementation of the subsequent action plan. The completion of the revised FF tool / implementation of the action plan is captured within the council's Corporate Procurement strategy 2016 – 2018; supporting Strategy Performance Action Plan. Revised Community Benefit Policy In our continued drive to embed wider sustainable procurement across the council, colleagues from Development and Regeneration Services (DRS) and the CPU formed a working group to develop and expand the council's Community Benefit Policy (CBP) to assist in enabling wider and more innovative Community Benefits outcomes for the council in terms of sustainability, corporate social responsibility and the Single Outcome Agreement. The policy was expanded to ensure Community Benefits are considered and included in all regulated procurement including all relevant contracts i.e. works (£500k), goods and services contracts (£50k). To gain full benefit of the revised policy, the CPU developed a Community Benefits Tender Evaluation and Reporting Model which will enhance the opportunity of suppliers' proposals and achieved outcomes aligning to the council's priority areas and will provide added value. Historically the Community Benefit tender evaluation model and clauses were predominately utilised in works contracts and centred on targeted training and recruitment. The revised policy and supporting tender evaluation model now targets wider community benefits covering a number outcomes: Vocational Training, Supply Chain Development, Community and Environmental Initiatives, Supported Business, Third Sector and Voluntary Initiatives, Social Enterprise and Co-operative Initiatives, Equality and Diversity Initiatives, Poverty Initiatives. The revised policy forms a critical component in our collaborative approach to complying with the Sustainable Procurement Duty and has been incorporated as an appendix within the Corporate Procurement strategy 2016 – 2018. A Strategic Procurement Training Methodology and Training Plan was devised to support and ensure all Corporate Procurement staff involved in strategic projects undertook relevant procurement training, including any training specific to embedding sustainability. This resulted in all strategic CPU staff undertaking training on the Marrakech Task Force approach, which has ensued in a more comprehensive approach to the development and implementation of sustainable public procurement. CPU staff, alongside client / project Leads and Service Department delegates participated in a workshop which supported the completion of the prioritisation tool. Supporting SMEs and Supported Business Increasing our Quick Quote (QQ) threshold from £20k to £50k - This assisted in making it easier for smaller businesses to engage and contract with the council, providing consistency, reducing bureaucracy and affording opportunities that were previously less accessible and thus supporting economic growth. The CPU have continued to support a number of Meet the Buyer / Supplier Events, as facilitated by Supplier Development Programme (SDP) and Scotland Excel and also engage regularly with Ready for Business, Glasgow Social Enterprise Network (GSEN), and the British Association for supported employment (BASE) to assist in understanding their organisations needs and share the council's procurement work plan. To ensure Best value and support council wide policies: This is achieved by the application of procurement best practise and adopting the Scottish Model of Procurement (Price versus Quality versus Sustainability): applying M.E.A.T criteria

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which includes a quality sub-criteria for Community Benefits (10%) and Fair Work Practise (5%) in all relevant tenders. Undertaking key strategic activities: early client and market engagement, market research and analysis to assist in developing appropriate procurement strategies (e.g. small lots).Following our robust contract management process to ensure contracts provide best value i.e. monitor achieved performance, seek additional sustainable opportunities, added value / cost reductions, alternative solutions, voluntary community benefits etc.

Our strategic procurement process, procurement toolkit and various procurement documentation ensures that council policies are considered and supported when undertaking procurement activities. The majority are embedded within our strategic procurement process and procedures e.g. Sustainable Policy 2010, CSR, H&S, payment of the Living wage, Equalities and Carbon Management policy, Community Benefits , council signed up to the supplier CHARTER etc.

5(b) How has procurement activity contributed to compliance with climate change duties?
Provide information relating to how procurement activity by the body has contributed to its compliance with climate changes duties.
All procurement is undertaken in line with the various procurement rules and regulations, this includes the utilisation of the various Scottish Government tools and applications i.e. sPQQ, PCS-T, PCS etc. The Glasgow City Region City Deal Procurement Strategy incorporates key Strategic Objectives in relation to Sustainability, the aim being that the Legal and Procurement Support Group will deliver sustainability through procurement by ensuring that we consider social, ethical, environmental and economic impacts in our procurement activity. Significant actions/projects supported by the Council's procurement teams include: Tender: Supply of Horticultural / Ground Maintenance Materials, Trees, Plants & Seeds. The tender stipulated that the supply of goods or products must not cause significant damage to the environment. As part of the Sustainable Procurement Duty, Tenderers were asked how they will provide sustainability benefits across5 key areas: Sustainability, Water Consumption, Waste / Recycling, Hazardous, materials , Transportation / Deliveries. Many of the products are raw materials but it was recognised that pesticides are still used in the growing process and suppliers shall adhere to legislation and minimise or restrict the use of chemicals so as not to harm the environment. Suppliers must actively ensure that they manage water consumption by reuse of rain water and installation of irrigation systems. Waste / recycling of products and packaging are managed efficiently. Suppliers are now collecting packaging for reuse or are donating it to local communities which are in line with The Department for Environment, Food and Rural Affairs (DEFRA) recommendations. Transportation plays a part in the delivery of this framework and suppliers have recognised that this is key to ensuring products are delivered on time whilst reducing emissions in the environment. Suppliers recognise the importance of maximising resource efficiency and relevant opportunities for circular economy outcomes (make, use, dispose in order to keep resources in use for as long as possible) e.g. a supplier reuses shoe boxes sourced from a local supplier to send out some small seed orders and 2nd life combine harvesters once retired as research stations. This activity is key to this contract and suppliers will do their utmost to maintain these practices which shall be tracked through CPU contract management process. Tender: The Provision of a Framework Agreement for Taxi and Private Hire Services. Environmental Impacts: All companies have confirmed that they have an Environmental Policy. Two companies confirmed that they have BS EN ISO 14001 Environmental Accreditation. To obtain BS EN ISO14001 accreditation both companies have established an Environmental Management System within their organisation that enables them to enhance their environmental performance. The System meets with International Standards for Environmental Management and is assessed by an external certification body with certification being conditional on maintaining performance standards throughout the period of the certificate. The certification body carry out audits to ensure that the company maintain the performance of their Environmental Management System. All companies agreed to invoice electronically and thus eliminate the use of paper invoices. Companies have invested in newer systems and technology and are able to improve on route planning using GPRS and more easily dispatch the closest appropriate vehicle to the job. This helps to reduce mileage, fuel consumption and carbon emissions. Vehicles are replaced within 7 years and as companies procure new vehicles some of them are committing to reducing engine sizes and buying vehicles with start/stop facility. All vehicles are subject to bi-annual checks including checks on emission levels by the Council's Inspection Centre. Tender: Winter Maintenance Emergency Assistance. We included sustainable evaluation questions that specifically targeting suppliers environmental policies in relation to the reduction of vehicle emissions in the provision of our service requirements. We have also introduced a Cycle to Work staff Salary Sacrifice scheme. Current Contracts: Carriageway Thin Surface Treatment Works - Carriageway thin surface treatments are increasingly used and this reduces the need to use conventional surface treatments. Conventional surface treatments are hot applied treatment using fuel and energy as part of the heating process. Thin surface treatments are cold mix materials offering Organic Waste - short term contract in place for Garden Waste from May 2015 – December 2015, superseded by a contract for our full Organic Waste requirements which is Garden and Food Waste. Food Containers for Waste - To support the Waste (Scotland) Regulations2012 which states that Local Authorities should be collecting food waste by January 2016 and thus avoiding food waste going to landfill. GCC awarded a contract for the purchase of food caddies, back court bins and bio-degradable bags which is now being implemented and distributed to 300,000 households in Glasgow. Bulky Waste - the average round trip distance of less than 70 miles is the same as with the Organic Waste and for the same sustainable reasons. Suppliers on this contract must meet the recycling target of a minimum of 60% on a monthly basis to avoid bulky waste going to landfill. Pay As You Go Car Club - In the provision of the Pay As You Go Car Club it was stipulated that"the use of cleaner fuel vehicles is preferred e.g. fuel-cell, full electric, petrol electric hybrid or duel fuel (LPG with petrol). Prohibited vehicles include diesel vehicles, including diesel/LPG vehicles." Cremators – this contract included the purchase of 6 FULLY ABATED cremators which has afforded the Council the opportunity to secure the most efficient, environmentally responsible and technologically advanced cremation equipment. In installing 6 fully abated cremators Glasgow City Council will exceed its legal requirement under the PG5/2(12) legislation to abate 50%

5(c) Supporting information and best practice
Provide any other relevant supporting information and any examples of best practice by the body in relation to procurement.
It was recognised that a more robust and detailed process was required for procurement workload analysis and planning. Key to this process was establishing a framework of average hours for procurement activities which, in turn, was used to establish the capacity needed to deliver the procurement work plans and ensure our resources are allocated on an efficient and effective manner. This robust resource management workload planning process has assisted in ensuring the on-going delivery of many strategic and high risk projects within the council whilst also providing informative management information which ascertains the productivity within the procurement teams, resource requirements to sustain the work plans for the various clients / service departments. This framework is supported by time tracking for individual procurement team members to monitor and validate the average hours assigned. In the interest of collaboration and sharing best practise, the procurement workload analysis and planning framework was shared with the procurement sections of the other 31 Local Authorities. To support and ensure sustainable procurement, the CPU revised the client's 'initial procurement assessment' (IPA) document to include the link to the revised Community Benefits Policy and also the outcomes of the sustainable test. The IPA must be submitted as part of the client instruction prior to any procurement activity commencing. Many actions in relation to Sustainable Procurement and responding to the Climate Change Duties are taken forward by GCC Services and its partner organisations. This includes the following: PC Shutdown software and Intelligent Heating Controls (i.e. Building Management Systems or BMS) have been purchased to ensure energy use is minimized. Re-use software asset re-use software has been introduced on a trial basis to encourage re-use of surplus assets and thus minimise unnecessary procurement and the associated carbon.

PART 6: VALIDATION AND DECLARATION

6(a) Internal validation process

Briefly describe the body's internal validation process, if any, of the data or information contained within this report.

The data/information contained within this report was reviewed by officers independent to the teams involved in climate change mitigation and adaptation including the body's own Internal Audit Team and Land and the Environmental Services (LES) Performance Officer. We ensured that any supporting evidence was accurately referenced in the submission and that statistics were arithmetically accurate. The scope of the independent validation did not include an assessment of the appropriateness of the supporting documentation referenced in relation to the questions posed in the return. However, both the responses and the associated supporting documentation was referred to appropriate officers in LES with relevant knowledge and expertise for their review. Following the verification exercise, the parties involved concluded that the respective areas they verified, did, in the main provide a fair representation of the body's current performance in relation climate change.

6(b) Peer validation process

Briefly describe the body's peer validation process, if any, of the data or information contained within this report.

No Peer Validation took place.

6(c) External validation process

Briefly describe the body's external validation process, if any, of the data or information contained within this report.

No external Validation took place.

6(d) No validation process

If any information provided in this report has not been validated, identify the information in question and explain why it has not been validated.

The data/information contained within this report was reviewed by officers independent to the teams involved in climate change mitigation and adaptation including the body's own Internal Audit Team and Land and the Environmental Services (LES) Performance Officer. We ensured that any supporting evidence was accurately referenced in the submission and that statistics were arithmetically accurate. The scope of the independent validation did not include an assessment of the appropriateness of the supporting documentation referenced in relation to the questions posed in the return. However, both the responses and the associated supporting documentation was referred to appropriate officers in LES with relevant knowledge and expertise for their review. Following the verification exercise, the parties involved concluded that the respective areas they verified, did, in the main provide a fair representation of the body's current performance in relation climate change.

6e - Declaration		
I confirm that the information in this report is accurate and provides a fair representation of the body's performance in relation to climate change.		
Name	Role in the body	Date
George Gillespie	Assistant Director of Land and Environmental Services	2016-11-30

RECOMMENDED – WIDER INFLUENCE

Q1 Historic Emissions (Local Authorities only)

Please indicate emission amounts and unit of measurement (e.g. tCO2e) and years. Please provide information on the following components using data from the links provided below. Please use (1) as the default unless targets and actions relate to (2).

(1) UK local and regional CO2 emissions: **subset dataset** (emissions within the scope of influence of local authorities):

(2) UK local and regional CO2 emissions: **full dataset**:

Select the default target dataset

Full

Table 1a														
Source	Dataset	Sector	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Units	Comments
Full	DECC Sectors	Total Emissions	3770.13	3776.45	3681.12	3766.83	3232.65	3416.67	3133.90	3344.99	3176.04	2657.05	ktCO2	
		Industry and Commercial	1762.90	1793.32	1741.44	1846.47	1484.39	1608.79	1496.31	1642.57	1533.13	1203.69	ktCO2	
		Domestic	1407.49	1393.22	1343.56	1345.75	1192.74	1261.19	1106.87	1179.97	1129.38	933.55	ktCO2	
		Transport total	599.74	589.90	596.12	574.60	555.52	546.69	530.72	522.45	513.53	519.80	ktCO2	
		Per Capita	6.62	6.64	6.44	6.54	5.56	5.83	5.28	5.62	5.32	4.43	tCO2	
	Other Sectors	Waste											tCO2e	
		LULUCF Net Emissions											ktCO2	
		Other (specify in 'Comments')												

Table 1b														
Source	Dataset	Sector	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Units	Comments
Full	DECC Sectors	Total Emissions	4096.20	4104.26	4012.52	4091.84	3547.29	3730.49	3453.06	3675.70	3509.91	2983.46	ktCO2	
		Industry and Commercial	1765.74	1795.99	1744.24	1849.08	1485.82	1610.72	1497.34	1643.43	1534.34	1204.71	ktCO2	
		Domestic	1407.49	1393.22	1343.56	1345.75	1192.74	1261.19	1106.87	1179.97	1129.38	933.55	ktCO2	
		Transport total	922.97	915.05	924.72	897.00	868.73	858.58	848.86	852.30	846.19	845.20	ktCO2	
		Per Capita	7.23	7.25	7.05	7.13	6.13	6.39	5.85	6.20	5.91	5.00	tCO2	
	Other Sectors	Waste											tCO2e	
		LULUCF Net Emissions	20.07	18.91	18.21	17.28	16.55	15.72	14.70	14.12	13.47	12.92	ktCO2	
		Other (specify in 'Comments')												

Table 1c														
Source	Dataset	Sector	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	Units	Comments
Other	DECC Sectors	Total Emissions												
		Industry and Commercial												
		Domestic												
		Transport total												
		Per Capita												
	Other Sectors	Waste											ktCO2	
		LULUCF Net Emissions												
		Other (specify in 'Comments')												

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Q2a – Targets										
Please detail your wider influence targets										
RPP Sector	Action Type	Description	Type of Target (units)	Baseline value	Start year	Target saving	Target / End Year	Saving in latest year measured	Latest Year Measured	Comments
Energy	District Heating									
Transport	Decarbonising Private Vehicles									
Transport	Decarbonising Public Transport									
Transport	Modal Shift - Private Vehicle to Active Travel (Cycling/Walking)									
Homes and Communiti es	Energy Company Obligation (ECO)									

Q2b) Does the Organisation have an overall mission statement, strategies, plans or policies outlining ambition to influence emissions beyond your corporate boundaries? If so, please detail this in the box below.

The aims of the Sustainable Glasgow partnership, and developments in its work, are set out under Governance (required section). During 2014-15 it has supported the Area-wide actions listed above. In particular, the new Energy & Carbon Masterplan. This was done through the participation of the Council and partners in STEP-UP, a sustainable energy research and knowledge exchange project spanning northern Europe.

Sustainable Glasgow is the key partnership for delivering on city-wide sustainability aims, bringing together public bodies, the private sector and academia. It is chaired by the Leader of the Council, thereby ensuring the Council's strong leadership role in progressing the mitigation and adaptation agendas. The partnership has driven a number of green energy initiatives with a focus on affordable warmth and emissions reduction targets. This work has aimed to deliver major investment, help regenerate communities and tackle fuel poverty, and improve environmental quality. (see 'Required section) The Sustainable glasgow Initiative aims to achieve a 30% reduction in CO2 by 2020 relative to 2006-07 baseline, supporting the Scottish Government's 42% reduction target. This is further endorsed by the Energy and Carbon Masterplan.

Q3) Policies and Actions to Reduce Emissions																	
RPP Sector	Action Type	Description	Start year for policy / action imple - mentation	Year that the policy / action will be fully imple - mented	Annual CO2 saving once fully imple - mented (tCO2)	Latest Year measured	Saving in latest year measured (tCO2)	Status	Metric / indicators for monitoring progress	Delivery Role	During project / policy design and implementation, has ISM or an equivalent behaviour change tool been used?	Please give further details of this behaviour change activity	Value of Investment (£)	Ongoing Costs (£/ year)	Primary Funding Source for Implementation of Policy / Action	Accountable body	Comments
Energy	Other (please specify in comments)	Energy & Carbon Masterplan	2015														

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Q3) Policies and Actions to Reduce Emissions																	
RPP Sector	Action Type	Description	Start year for policy / action imple - mentation	Year that the policy / action will be fully imple - mented	Annual CO2 saving once fully imple - mented (tCO2)	Latest Year measured	Saving in latest year measured (tCO2)	Status	Metric / indicators for monitoring progress	Delivery Role	During project / policy design and implementation, has ISM or an equivalent behaviour change tool been used?	Please give further details of this behaviour change activity	Value of Investment (£)	Ongoing Costs (£/ year)	Primary Funding Source for Implementation of Policy / Action	Accountable body	Comments
Energy	District Heating	Creation of an Energy Services Company (ESCo), focused mainly on delivering district heating and renewables. District heating projects: • Glasgow Recycling & Renewable Energy Centre (at Polmadie) • Glasgow University DH extension • University of Strathclyde DH (as part of City Centre North	2014														
Energy	Renewables - Solar Farm	TSB Energy Demonstrator: • Mapping of photovoltaic potential on vacant & derelict land, to encourage PV schemes (business, community or Council-led). • Online Virtual Building Energy Model; • Demand Side Management in operation in selected GCC buildings	2014														
Homes and Communiti es	Behavioural	TSB Energy Demonstrator - Behavioural Change: implement community engagement on energy management, based on 2013-14 Glasgow University study. Housing and energy efficiency programme to continue through HEEPS-ABs funding. Sighthill Transformation Regeneration Area (TRA), insulation of the GHA stock and installation of district heating scheme. Glasgow Green Year 2015.	2015														

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Q3) Policies and Actions to Reduce Emissions																	
RPP Sector	Action Type	Description	Start year for policy / action imple - mentation	Year that the policy / action will be fully imple - mented	Annual CO2 saving once fully imple - mented (tCO2)	Latest Year measured	Saving in latest year measured (tCO2)	Status	Metric / indicators for monitoring progress	Delivery Role	During project / policy design and implementation, has ISM or an equivalent behaviour change tool been used?	Please give further details of this behaviour change activity	Value of Investment (£)	Ongoing Costs (£/ year)	Primary Funding Source for Implementation of Policy / Action	Accountable body	Comments
Business Industry and Public Sector	Other (please specify in comments)	Engagement with the Chamber of Commerce to identify opportunities in this sector through its Green Business Network.	2014														
Transport	Decarbonising Public Transport	Two electric buses currently service the Riverside 100 route.	2014														
Transport	Modal Shift - Private Vehicle to Active Travel (Cycling/Walking)	The City Bike Hire Scheme, with around 400 cycles at 30 automated hire stations in and near the city centre	2014														
Transport	Transport Planning	Opening of the Fastlink dedicated rapid bus route in 2015.	2016														
Transport	Modal Shift - Private Vehicle to Active Travel (Cycling/Walking)	Active travel apps project to start in operation.	2014														
Waste and Resource Efficiency	Energy from waste	Progress construction of Glasgow Recycling & Renewable Energy Centre (at Polmadie), including Energy Recovery Facility using the waste-derived fuel. The proposed South Clyde Energy Centre would similarly create fuel from waste and then derive energy from it. A commercial planning application was approved in 2012. The site is almost near completion and should be operating in the near future.	2014														

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Q3) Policies and Actions to Reduce Emissions																	
RPP Sector	Action Type	Description	Start year for policy / action imple - mentation	Year that the policy / action will be fully imple - mented	Annual CO2 saving once fully imple - mented (tCO2)	Latest Year measured	Saving in latest year measured (tCO2)	Status	Metric / indicators for monitoring progress	Delivery Role	During project / policy design and implementation, has ISM or an equivalent behaviour change tool been used?	Please give further details of this behaviour change activity	Value of Investment (£)	Ongoing Costs (£/ year)	Primary Funding Source for Implementation of Policy / Action	Accountable body	Comments
Rural Land Use	Other (please specify in comments)	outputs from iTree assessment project to integrate quantified C sequestration benefits with strategy for creation and management of City woodlands. Begin priority enhancements for non-Council-owned wetlands in Seven Lochs Wetland Park, securing and/or enhancing carbon sequestration - working with GCV Green Network Partnership, anticipating SNH Peatland Action funding.	2015														

Please provide any detail on data sources or limitations relating to the information provided in Table 3

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Q4) Partnership Working, Communication and Capacity Building. Please detail your Climate Change Partnership, Communication or Capacity Building Initiatives below.										
Key Action Title	Action Type	Organisation's project role	Lead Organisation (if not reporting organisation)	Private Partners	Public Partners	3rd Sector Partners	Outputs	Value to Organisation	Total Investment into Partnership	Comments
Research & Development	Partnership working of climate change or sustainability	Participant	Adaptation Scotland		GCVN; Clyde Plan; MGSDP					Climate Ready Clyde
Research & Development	Partnership working of climate change or sustainability	Lead			University of Strathclyde, GHA					Sustainable Glasgow
Investment	Partnership working of climate change or sustainability	Lead			SEPA, North Lanarkshire, South Lanarkshire, Scottish Water					MGSDP

OTHER NOTABLE REPORTABLE ACTIVITY

Q5) Please detail key actions relating to Food and Drink, Biodiversity, Water, Procurement and Resource Use in the table below.				
Key Action Title	Key Action Description	Organisation's Project Role	Impacts	Comments
Food & Drink				
Biodiversity	Local Biodiversity Action Plan	Lead	Refreshed LBAP drafted (refers to climate change and potential impacts on biodiversity in Glasgow).	LBAP broad habitats (Grasslands, Farmland, Wetlands, Woodlands): current targets are to ensure no net loss of habitat and attempt to increase habitat extent and quality. Priorities include wetland enhancements in the emerging Seven Lochs Wetland Park, more pond naturalisations, and further wildflower enhancements to grasslands. iTree assessment reports underpin management planning for Glasgow's 'urban forest'.
Biodiversity	Local Nature Reserve Vision prepared for 2015-2020	Lead	Local Nature Reserves (LNRs) are places that have a natural value in terms of the rarity or special qualities of the wildlife and/or the wildlife habitat that can be found there. For instance one (Commonhead Moss) is a raised bog, a very rare and threatened habitat in Glasgow, whilst others have rare species such as the purple hairstreak butterfly found in woodland habitat at Garscadden Woods and Dawsholm Park LNR's.	

Q6) Please use the text box below to detail further climate change related activity that is not noted elsewhere within this reporting template

• Open Space strategy
Glasgow City Council is currently developing a new Open Space Strategy that seeks to connect or creates a network of publicly usable multifunctional open spaces across Glasgow. Climate change, adaptation and integrated green infrastructure is a key priority for open space development and delivery. We are developing local contexts that provide scoping and background information for six Strategic Development Framework (SDF) areas – parts of the City that span beyond neighbourhood level. North Glasgow is one of the SDF areas. The Open Space Strategy will be published before the end of 2016.

• The Woodside, Firhill, and Hamitonhill MasterPlan has been developed following the 2015 local charrette.
Funding has now been secured from a range of stakeholders to undertake improvements in line with Master Plan opportunities. The Hamiltonhill Claypits Local Nature Reserve is a 6.73 hectare linear reserve located on former industrial land. Partners have ambitions to explore the potentials for developing an inner-city conservation project. Over the coming year, Friends of Possilpark Greenspace are seeking to engage partners and local communities in a conversation to define how environmental and accessibility improvements could be delivered. There is an opportunity for climate resilience to feature in these discussions so that improvements also serve as effective long-term climate solutions.

• Stalled spaces
The Stalled Spaces programme supports community groups and local organisations develop temporary projects on stalled sites or underutilised open spaces in Glasgow. Projects are community led and initiatives respond to the needs of the local area. In winter 2016/2017, we will commence a study with CB&I to evaluate and assess vacant and derelict land under 0.1hectare in the North Glasgow SDF. This will provide communities with relevant data to inform development of projects. There is an opportunity to consider long-term trends, such as climate change, as part of the assessment of these spaces.

• Weathering Change community engagement in North Glasgow

• Engagement with Scottish Canals/ canals college

• New City Development Plan
Recommendations from Scottish Government's examination of Glasgow's Proposed City Development Plan were provided in June 2016. Recommendations and output from public consultations will be addressed over the coming months with adoption of a final City Development Plan expected for January 2017. The City Development Plan contains a suite of Supplementary Guidance to support the plan in its development and delivery. We are currently looking to identify how resilience thinking can be built into the City Development Plan action plan. In particular, we want to consider the how climate change will impact on homes, the city centre, the rivers, city infrastructure and businesses.

• New Environmental Taskforce / maintenance issues.
The city's new Environmental Taskforce (ETF) was launched in May 2016. The ETF focuses on improving and making best use of environmental services from across the Glasgow Council Family and partner organisations, and delivering them at a local level. The launch of the ETF marks a change in the way the city is maintained – with the public encouraged to define how resources are deployed across the city. A major recruitment drive is also underway to see unemployed people of all ages trained up as ETF officers.

In 2017, we will engage the EFT in a conversation to define how the service can be developed in a way that makes it resilient to a changing climate. At the same time, we would like to understand how the EFT could play a role as ambassadors for climate resilience.

• Cooperative development movement, and the URBACT project could support further specific engagement/resourcing such as the connection routes mapping.

As part of Glasgow's participation in the URBACT III ERDF RESILIENT EUROPE project, Glasgow is working with stakeholders in the Possilpark area to develop a Local Action Group (LAG). The LAG will develop an integrated action plan for building local resilience. Outcomes of acute poverty and deprivation have been identified as the greatest challenges for this neighbourhood. Poor connectivity and healthy eating options are two themes that have featured heavily in local discussions. Between September 2016 and June 2017, we will work with local community groups to:

- Explore the local challenges associated with urban food resilience and long-term security.
- Map connectivity and local access routes to determine barriers and opportunities.Assess options that engage diverse communities with growing opportunities.

Long-term stresses including climate change will be considered as part of this programme of engagement.